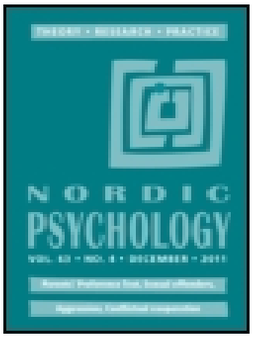

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RESEARCH ARTICLE

Trusting behavior and depressive symptoms

ELVAR FRIÐRIKSSON[1,2], HAUKUR FREYR GYLFASON[3], VAKA VÉSTEINSDÓTTIR[4] & JON FRIDRIK SIGURDSSON[2]

Correspondence address: Haukur Freyr Gylfason, Department of Business, Reykjavik University, Reykjavik, Iceland.
Email: haukurgy@ru.is

Abstract

Economic games have been shown to be an effective way to assess interpersonal impairment, which is associated with depression. Because trust is a key element in all social interaction, we tested whether trust, assessed with an economic game, is negatively associated with depressive symptoms. Participants (n = 163) took part in a distrust game and completed the Patient Health Questionnaire (PHQ-9), the Depression Anxiety and Stress Scales (DASS), and demographic questions. Multiple regression analysis estimated the relationship between the outcome of the distrust game and the cut-off scores of the PHQ-9, those of the DASS depression scale, and a depressive dichotomous score created using the cut-off scores from the PHQ-9 and the DASS depression subscale. The cut-off scores of the PHQ-9, the DASS depression scale, and the depressive dichotomous score separately predicted trust. The distrust game could be useful in further research into underlying mechanisms in interpersonal impairments and deficits in decision making in depression, which could lead to the development of interventions aimed at reducing the interpersonal impairments often associated with mental health.

Keywords: depression, trust, interpersonal processes, distrust game, decision making

Highlights

- We tested whether trust behavior is negatively associated with depressive symptoms.
- Trust behavior was assessed with an incentivized distrust game.
- The results indicate that depressed individuals show reduced trust.

Trust is considered a key element in all social interactions (Thielmann & Hilbig, 2015). It impacts interpersonal and group behavior (Buchan et al., 2008; Johansson-Stenman et al., 2005; Zak & Knack, 2001) and is situation specific (Fleeson & Leicht, 2006). Mayer et al. (1995) define trust as "the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party" (p. 712). The

[1]Department of Psychology, Reykjavik University, Reykjavik, Iceland
[2]Faculty of Medicine, University of Iceland, Reykjavik, Iceland
[3]Department of Business, Reykjavik University, Reykjavik, Iceland
[4]Department of Psychology, University of Iceland, Reykjavik, Iceland

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definition highlights how important trust is in interpersonal situations and how crucial vulnerability and the ability to infer other's behaviors are as elements of trust in others.

Different types of trust exist depending on the context (Simpson, 2012), although one type—i.e., trust among strangers—seems particularly relevant for the present study (McKnight et al., 1998). A variety of everyday social interactions require trusting strangers, thus excluding the long-term formation of a trusting relationship based on multiple interactions with that person or those persons (Bohnet & Zeckhauser, 2004; Dunning et al., 2012; Dunning & Fetchenhauer, 2010; Gill et al., 2005).

Individuals with mental illness present deficits in the ability to socialize (King-Casas & Chiu, 2012; Lam et al., 2003; Segrin, 2000) and cooperate (Cáceda et al., 2014; Clark et al., 2013; King-Casas et al., 2008; Ong et al., 2017). Diagnostic criteria specify that an inability to socialize is a factor in several disorders, such as social phobia, schizophrenia, autism, and borderline and antisocial personality disorders, and is associated with major depressive disorder (MDD) (DSM-5; American Psychiatric Association, 2013).

Trust in new encounters (i.e. strangers) is in some part reinforced through cognition-based processes, and therefore, contingent on cognitive cues and first impressions (McKnight et al., 1998). Individuals with depressive symptoms report more negative personal interactions, present excessive reassurance seeking (Hammen & Brennan, 2001; Joiner et al., 1999; Zlotnick et al., 2000), and demonstrate deficiency or failure in decision-making (Paulus, 2007). Although cognitive biases in depression have been studied intensively (Beck, 2008), few studies have examined how cognitive biases influence social economic decision making (such as in economic games), which could lead to poor outcomes for individuals inflicted with depression (Harlé et al., 2010).

To date, only a few economic games have been used in association with depressive symptoms (e.g. Clark et al., 2013; Ong et al., 2017; Unoka et al., 2009; Wang et al., 2014; Watabe et al., 2015; Zhang et al., 2012), with few clinical studies reporting that individuals with major depressive disorders appear to have some difficulties with social decision-making (Clark et al., 2013; Wang et al., 2014; Watabe et al., 2015; Zhang et al., 2012).

Three studies have reported on the association between trust among strangers, measured with an economic game, and depressive symptoms (Clark et al., 2013; Unoka et al., 2009; Watabe et al., 2015). (Unoka et al. compared clinical samples to healthy volunteers.) In all three, no association was found between depressive symptoms and trust measured with the trust game (Berg et al., 1995). However, in none of these studies did participants play against real people, as instructed in economic games. According to Vlaev (2012), real and hypothetical decisions trigger different cognitive biases, and people tend to overestimate their and others' tendency to act cooperatively if they do not experience real social interactions. Additionally, Ong et al. (2017) assert that past research indicates "that individuals with a history of mood disturbances may exhibit *increased* cooperative behavior when compared with healthy adults." (p. 1). People in general, and more so people experiencing mood disturbances, might therefore expect more reciprocity in studies of trust if they believed that they were not playing against a real person. Moreover, the trust game used in the previous studies is based in part on cooperation between the trustor and the trustee (Berg et al., 1995) and associated with altruism and risk aversion (McEvily et al., 2012; Sapienza et al., 2013) that could tangle the association between a behavioral measure of trust and depressive symptoms.

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Trust is a significant factor in mental health. Trust makes it possible for people to establish successful relationships, which would precede more socially supportive once (Watabe et al., 2015). Furthermore, trust reduces uncertainty and the feeling of diminished control in social decision-making situations (Unoka et al., 2009). And, since lack of control in social situations has been found to correlate with depressive symptoms (Griffin et al., 2002) we hypothesize that trust is negatively associated with more depressive symptoms, controlling for known correlates of depression such as age (Christensen et al., 1999), gender (Piccinelli & Wilkinson, 2000), and anxiety (Prinz et al., 2014).

The current study adds to previous studies by using multiple assessments of depressive symptoms, larger sample size, and a clear target of trust with participants playing against real people (Clark et al., 2013; Unoka et al., 2009; Watabe et al., 2015). But more importantly, by having participants play a different economic game measuring trust that is not reliant on cooperation, since in the game used in this study there is no opportunity to cooperate, only guard against harm (McEvily et al., 2012).

Method

Participants

A total of 197 undergraduate students participated in the study. However, 34 did not complete the distrust game[1], which left 163 students and their partners. In this study, the primary focus was on the student sample that replied to the mood questionnaires. Each student was paired anonymously with a passerby at a local shopping center. The student sample included 90 (55%) women and 73 (45%) men, and the average age was 23.14 ($SD = 4.66$) years. The study was approved by the National Bioethics Committee of Iceland (No. 15-163).

Measures and procedure

The distrust game. Students played a single round of the distrust game[2], a complementary behavioral measure of Berg et al. (1995) trust game (McEvily et al., 2012). The students had financial incentives to trust the passersby, who on the other hand had financial incentives to fleece the students, of which the students were informed. In very simple terms, from the viewpoint of the students, it is possible to see the distrust game as similar to purchasing an insurance against financial harm caused by the passersby.

Before the study, a PowerPoint presentation was given to the students in groups, providing instructions on how to play the distrust game. Subsequently, the students were asked to read the game's full instructions (McEvily et al., 2012).

The students were told that they would be playing against a real participant in a local shopping center ("passerby") and instructed that the pair had been allocated 1,000 ISK (500 ISK each) in total wealth.

The students were able to specify how much they were willing to pay, out of their allocated 500 ISK, not to have to play against a field participant (who would otherwise be able to take all, something or nothing of the student's allocated 500 ISK) and instead be guaranteed what was left of their 500 ISK. They could pay between 0 ISK and 500 ISK to cancel the field participant's discretion over the division of their 500 ISK.

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Table 1. An example of the decision sheet.

Amount you receive	Amount the other receives			Decision		Amount you receive	Amount the other receives
500	500	Pay 0	☒	1	☐	Play game	Determined by other person
450	500	Pay 50	☒	2	☐	Play game	Determined by other person
400	500	Pay 100	☒	3	☐	Play game	Determined by other person
350	500	Pay 150	☐	4	☒	Play game	Determined by other person
300	500	Pay 200	☐	5	☒	Play game	Determined by other person
…	…	…	☐	…	☒	…	…
…	…	…	☐	…	☒	…	…
…	…	…	☐	…	☒	…	…
0	500	Pay 500	☐	1	☒	Play game	Determined by other person

Note: Students filled out the Decision Sheet by checking one of two boxes for each decision (1 through 11). Each option was treated as an independent decision. Once the students had completed their Decision Sheet, an option was chosen at random by researchers to play against the passersby at the local mall. If decision 4 had been chosen at random, the student chose to play the game against the passerby, giving the passerby the option of taking all of the student's allotment, nothing or something in between. The student's expected return was then somewhere between 0-500 ISK. If decision 3 had been chosen at random, the student chose not to play against the passerby and instead paid 100 ISK to be able to keep 400 ISK.

The students were introduced to the Decision Sheet, a form that contained 11 decision options (Table 1), and asked to indicate how much they were willing to pay not to play the distrust game against the field participant. Each decision had two options, i.e., "pay X" to avoid playing the game or play the game. The "pay X" choice was in 50 ISK increments (e.g., decision one 0 ISK, decision two 50 ISK, … decision eleven 500 ISK). Students were told that one of the 11 options would be chosen at random and instructed to treat each decision option as if it were the only one that counted.

When the students had filled out the Decision Sheet, one of the 11 options was chosen at random (as conveyed to the students), and that decision was used as the student's move against the field participant. If the students had selected "pay X" not to play the game, then the student received ISK 500 - X. If the students had selected "play the game", then field participants chose how much of the students' share they wanted to keep for themselves (all of it, some of it, or none of it) and the students got the residual.

A composite measure for distrust was computed by summing the scores for each of the decisions (0 = pay not to play the game; 1 = to play the game). Scores ranged from 0 to 10, with a higher score indicating more trust.

During the following two weeks, after performing the data collection at the university, the researchers collected data from the field participants at a local shopping center. They were informed that they were playing against an undergraduate student at university and

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received their payment on site. The students received their earnings following the completion of the data collection at the shopping mall.

Mood and anxiety symptoms. After playing the distrust game, students completed the Icelandic version of the Patient Health Questionnaire (PHQ-9; Jónsdóttir & Sigurðardóttir, 2016; Kroenke et al., 2001; Spitzer et al., 1999), a nine-item self-report questionnaire. The four-point scale ranged from 0 (not at all) to 3 (nearly every day). Total score was calculated by summing its scores.

In their meta-analysis, however, Manea et al. (2012) recommended a cut-off score between 8 and 11 for the PHQ-9, and Kroenke and Spitzer (2002) recommended a cut-off score of 10 for the PHQ-9 to detect a current major depressive episode. According to Kroenke et al. (2001), the four cut-off points of 5, 10, 15, and 20 characterize mild, moderate, moderately severe, and severe depression, respectively. We chose a cut-off score of 10 in this study, a cut-off score that represents moderate depression (Kroenke et al., 2001).

The students also completed the Icelandic version of the Depression Anxiety and Stress Scales (DASS; Ingimarsson, 2010; Lovibond & Lovibond, 1995b), a 42-item self-report questionnaire that requires participants to rate how accurately the statements describe their behavior, emotions or thoughts for the past week. The four-point scale ranged from 0 (did not apply to me at all) to 3 (applied to me very much, or most of the time). The DASS has three subscales, measuring depressive, anxiety and stress symptoms. Total scores were calculated by summing the scores for the relevant 14 items.

Lovibond and Lovibond (1995a) divided each DASS subscale into five categories (normal, mild, moderate, severe and extremely severe) to use as cut-off scores. Based on Z-scores, a score less than 0.5 was considered normal; 0.5 to 1.0 was considered mild; 1.0 to 2.0 was considered moderate; 2.0 to 3.0 was considered severe; and greater than 3.0 was considered extremely severe. We chose a cut of score of 13 in this study (corresponding to a Z-score of 1.0), a cut-off score that represent moderate depression (Lovibond & Lovibond, 1995a) and is consistent with a cut-off score of 10 on the PHQ-9 (Kroenke et al., 2001).

Results

For the DASS, the average score was 6.14 ($SD = 6.77$) for the Depression subscale, 5.55 ($SD = 5.96$) for the Anxiety subscale, and 9.36 ($SD = 7.93$) for the Stress subscale. For the PHQ-9, the average score was 5.61 ($SD = 4.18$), and for the distrust game the average was 3.34 ($SD = 2.26$). Descriptive statistics for the PHQ-9 and the DASS cut-off scores can be seen in Table 2.

A measure of a depressive dichotomous score was created using the cut-off scores of both the PHQ-9 and the DASS Depression subscale. Participants were coded as depressed if they scored within both depression scales' cut-off scores. If participants scored below the cut-off scores on either one or both scales, they were coded as being not depressed. The average score on the PHQ-9 for participants who were coded depressed was considerably higher compared to participants coded not depressed, $t(161) = 12.66$, $p < .001$ (Table 2). The same was true for the DASS Depression scale, $t(161) = 16.22$, $p < .001$.

There was a negative association between the DASS Depression subscale ($r_p = -.10$, $p = .06$), PHQ-9 cut-off score ($r_p = -.12$, $p = .05$), and the Depressive dichotomous score ($r_p = -.18$, $p = .01$) with trust (p-values are one-tailed analogous to hypothesis).

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Table 2. Means and standard deviations for DASS depression subscale, DASS anxiety subscale, PHQ-9, and the trust measure for groups (not depressed and depressed) by cutoff scores for PHQ-9, DASS depression scale, and the depressive dichotomous score.

			M	SD	N	%
PHQ-9 (≥ 10)						
	DASS depression subscale					
		Not depressed	3.83	3.89	135	82.82
		Depressed	17.29	6.68	28	17.18
	DASS anxiety subscale					
		Not depressed	4.04	4.38	135	82.82
		Depressed	12.82	7.17	28	17.18
	PHQ-9					
		Not depressed	4.09	2.51	135	82.82
		Depressed	12.96	2.43	28	17.18
	Trust					
		Not depressed	3.44	2.27	135	82.82
		Depressed	2.82	2.20	28	17.18
DASS depression scale (≥ 13)						
	DASS depression subscale					
		Not depressed	3.33	2.92	132	80.98
		Depressed	18.13	5.21	31	19.02
	DASS anxiety subscale					
		Not depressed	3.98	4.03	132	80.98
		Depressed	12.23	7.99	31	19.02
	PHQ-9					
		Not depressed	4.26	3.02	132	80.98
		Depressed	11.39	3.45	31	19.02
	Trust					
		Not depressed	3.47	2.30	132	80.98
		Depressed	2.77	2.05	31	19.02
PHQ-9 (≥ 10) and DASS depression scale (≥ 13)						
	DASS depression subscale					
		Not depressed	3.99	3.93	140	85.89
		Depressed	19.26	5.54	23	14.11
	DASS anxiety subscale					
		Not depressed	4.33	4.69	140	85.89
		Depressed	13.00	7.42	23	14.11
	PHQ-9					
		Not depressed	4.42	3.04	140	85.89
		Depressed	12.87	2.44	23	14.11
	Trust					
		Not depressed	3.50	2.27	140	85.89
		Depressed	2.35	2.04	23	14.11

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Table 3. Linear regression results for cut-off scores of the DASS depression scale and the PHQ-9 as predictors of trust (p-values are two-tailed).

	Trust Model 1				Trust Model 2				Trust Model 3			
	b	*SE*	95% CI	*p*	*b*	*SE*	95% CI	*p*	*b*	*SE*	95% CI	*p*
DASS anxiety	0.05	0.04	[-0.00, 0.13]	.147	0.06	0.04	[-0.02, 0.13]	.120	0.07	0.03	[-0.00, 0.13]	.059
Female	0.34	0.36	[-0.36, 1.04]	.340	0.29	0.36	[-0.41, 0.99]	.417	0.28	0.35	[-0.41, 0.97]	.425
Age (centered at 19 yrs.)	−0.08	0.04	[-0.15, 0.00]	.051	−0.08	0.04	[-0.15, −0.00]	.044	−0.08	0.04	[-0.15, −0.00]	.043
PHQ-9 (≥10)	−1.17	0.56	[-2.28, −0.07]	.038								
DASS depression (≥13)					−1.24	0.53	[-2.27, −0.19]	.021				
Depressive dichotomous score									−1.84	0.58	[-2.98, 0.71]	.002
R^2	.058				.064				.091			

Note. CI = confidence interval.

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A linear regression was used to determine whether the cut-off scores on the PHQ-9 ($\geq$ 10), the DASS Depression scale ($\geq$ 13), and the dichotomous score predicted the outcome of the distrust game (trust) while controlling for gender, age and anxiety (Table 3). The independent variables were entered simultaneously after it was ascertained that no assumptions were violated, including the assumption of multicollinearity. The cut-off score for the PHQ-9 (model 1), that of the DASS Depression scale (model 2), and that of the dichotomous score (model 3) all significantly predicted less trust, holding the effects of all other predictors constant.

Discussion

The purpose of the present study was to examine whether depressive symptoms were negatively associated with trust. Our results reveal a negative association between trust and depressive symptoms, not consistent with previous results (Clark et al., 2013; Unoka et al., 2009; Watabe et al., 2015). However, we used real counterparts as instructed in economic games, which was not the case in previous studies (Clark et al., 2013; Unoka et al., 2009; Watabe et al., 2015), and might have reduced the risk of our participants, and depressed individuals in particular, to overestimate other's tendencies to act cooperatively (Ong et al., 2017). Additionally, our sample was more than twice the size of that of Watabe et al. (2015), three times the size of Unoka et al. (2009), and four times the size of Clark et al. (2013).

The measure of depressive symptoms was performed using self-report scales, which are different from an actual diagnosis of major depressive disorder given by a clinician. Moreover, the study assessed depressive symptoms in a sub-clinical student population. Recruiting participants with a diagnosis of major depressive disorder as an experimental group and assessing gameplay compared to a non-clinical sample would have been desirable. However, by using multiple assessments of depressive symptoms and recognized cut-off scores, we believe we came that much closer to being able to correctly identify students experiencing depression. In fact, 43.5% of those classified as being depressed scored more than 2 Z-scores above the mean on the DASS depression scale, indicating severe or extremely severe symptoms of depression (Lovibond & Lovibond, 1995a). Additionally, the average score on the DASS Depressions scale and the PHQ-9 for participants who were more likely to be diagnosed with depression mirrored that of clinical samples (Antony et al., 1998; Beard et al., 2016; Brown et al., 1997; Cunningham et al., 2016; Fracalanza et al., 2014).

Furthermore, the part of the distrust game the students played is rather challenging. In total, 34 students were unable to complete the game, a rate not uncommon in similar methodological studies (Charness et al., 2013). However, unlike the trust game, the distrust game does not have elements of cooperation and the same amount of altruism or risk aversion (McEvily et al., 2012; Sapienza et al., 2013) that could tangle the association between a behavioral measure of trust and depressive symptoms.

Moreover, given that the study was cross-sectional, inferences on the temporal relationship between depressive symptoms and trust cannot be made. We followed Unoka et al. (2009) and Watabe et al. (2015) and regressed depressive symptoms on trust. However,

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Clark et al. (2013) regressed trust on depressive symptoms. Not surprisingly, there is the possibility that the relationship is bidirectional, which requires further investigation.

This study was the first that we know of to show a relationship between depressive symptoms and a behavioral measure of trust measured in an economic game. Economic games have been shown to be an effective way to assess interpersonal impairments, which is associated with depression (Clark et al., 2013; Segrin, 2000; Watabe et al., 2015).

We know that the ability to trust is important in all social interactions (Thielmann & Hilbig, 2015) and that depressed individuals have difficulties in trusting others (Kupferberg et al., 2016). We also know that depressed individuals seem to have impaired ability to process signals of trust and translate them appropriately into behavior (Clark et al., 2013; Cusi et al., 2013). Future research should look at training depressed individuals to translate trust into behavior using behavioral experiments.

We agree with others (e.g. Anderl et al., 2018; Clark et al., 2013) that future research should uncover and isolate underlying mechanisms that drive interpersonal impairments and deficits in decision making, which economic games such as the distrust game measure, and discover how they may be related to specific disorders.

Notes

1. There was no difference between participants who did not complete the distrust game and participants who completed the distrust game on gender, age, DASS depression subscale, DASS anxiety subscale or PHQ-9.
2. McEvily's et al. (2012) Game Instructions for the distrust game can be found at http://www.sciencedirect.com/sdfe/arp/media/1-s2.0-S0899825611001102-mmc1.pdf

Disclosure statement

No potential conflict of interest was reported by the authors.

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